

Simulation Lab Experiments

1. Simulation of Single Server Single Queueing System

Aim

To simulate a single server single queueing system using GPSS and study customer waiting time, queue length, and server utilization.

Theory

A single server queueing system has one server and one waiting queue. Customers arrive randomly and wait if the server is busy. The system follows the First Come First Serve rule. Simulation helps analyze queue length, waiting time, and server utilization.

Procedure

1. Create the GPSS model for customer arrivals using the GENERATE block.
2. Use the QUEUE block to form the waiting line.
3. Use SEIZE to allocate the barber/server.
4. Simulate service time using ADVANCE.
5. Release the server using RELEASE.
6. Terminate transactions after service completion.
7. Run the simulation for 480 minutes.
8. Observe queue length, waiting time, and utilization.

Conclusion

The simulation demonstrates the behavior of a single server queueing system and helps analyze service efficiency, customer waiting time, and server utilization.

2. Simulation of Multi Server Single Queueing System

Aim

To simulate a multi-server single queueing system and evaluate system performance with multiple servers.

Theory

A multi-server queueing system contains multiple servers connected to a common queue. Customers are served by the first available server. This system reduces customer waiting time and improves service efficiency. It is commonly used in banks, airports, and hospitals.

Procedure

1. Generate customer arrivals using the GENERATE block.
2. Create a common queue using QUEUE.

3. Allocate available servers using SEIZE.
4. Simulate service using ADVANCE.
5. Release the servers using RELEASE.
6. Terminate completed transactions.
7. Run the simulation for 480 minutes.
8. Analyze queue statistics and server utilization.

Conclusion

The multi-server system reduces waiting time and improves service efficiency compared to a single-server model.

3. Simulation of Inventory System

Aim

To simulate an inventory management system and study inventory handling, customer service, and restocking operations.

Theory

An inventory system manages the storage and movement of goods in an organization. It helps maintain proper stock levels and avoid shortages or overstocking. Simulation is used to study worker utilization, customer demand, and restocking processes. It improves inventory control and operational efficiency.

Procedure

1. Generate customer demand transactions.
2. Place customers in a queue using QUEUE.
3. Allocate workers using SEIZE.
4. Simulate service time using ADVANCE.
5. Release workers using RELEASE.
6. Add a restocking process using GENERATE.
7. Stop the simulation after 480 minutes.
8. Analyze inventory flow and worker utilization.

Conclusion

The simulation helps evaluate inventory operations, worker efficiency, and customer waiting time for better inventory management.

4. Simulation of Quality Control System

Aim

To simulate a quality control system and analyze inspection efficiency and worker utilization.

Theory

A quality control system checks products to ensure they meet required quality standards. Products are inspected and classified as accepted or rejected. Simulation helps study inspection time, queue length, and worker utilization. It improves product quality and reduces delays in production.

Procedure

1. Generate products for inspection.
2. Add products to the inspection queue.
3. Allocate QC workers using SEIZE.
4. Simulate inspection time using ADVANCE.
5. Release workers after inspection.
6. Include a restocking process if required.
7. Run the simulation for 480 minutes.
8. Observe queue statistics and worker utilization.

Conclusion

The simulation shows how a quality control process can be analyzed to improve inspection efficiency and reduce delays.

5. Simulation of Washing Machine System

Aim

To simulate a washing machine system and study washing, rinsing, and spinning operations.

Theory

A washing machine system processes clothes through washing, rinsing, and spinning stages. Washing machines act as servers that handle customer loads. If machines are busy, clothes wait in queues for processing. Simulation helps analyze machine utilization and customer waiting time.

Procedure

1. Generate clothes arrival transactions.
2. Create washing, rinsing, and spinning queues.
3. Allocate washing machines using SEIZE.
4. Simulate washing, rinsing, and spinning using ADVANCE.
5. Release machines after each stage.
6. Terminate transactions after completion.
7. Run the simulation for 480 minutes.
8. Analyze queue length and machine utilization.

Conclusion

The simulation helps identify bottlenecks and evaluate machine utilization in a laundry service system.

6. Simulation of Airline Passenger Processing System

Aim

To simulate passenger processing at an airport and analyze queue behavior and resource utilization.

Theory

An airline processing system manages passenger movement through check-in, security, and boarding stages. Passengers wait in queues when service counters are busy. Simulation helps analyze queue length, waiting time, and facility utilization. It helps improve airport efficiency and passenger flow.

Procedure

1. Generate passenger arrivals.
2. Create queues for check-in and security.
3. Allocate resources using SEIZE.
4. Simulate service times using ADVANCE.
5. Release resources after service.
6. Simulate boarding movement.
7. Run the simulation for 480 minutes.
8. Observe queue lengths and facility utilization.

Conclusion

The simulation helps analyze passenger flow and identify improvements in airport operations.

7. Simulation of Supermarket System

Aim

To simulate a supermarket billing system and study billing counter utilization and customer waiting time.

Theory

A supermarket billing system handles customers arriving for payment services. Customers wait in a queue until the billing counter becomes available. Simulation helps study billing time, customer waiting time, and counter utilization. It helps improve customer service efficiency.

Procedure

1. Generate customer arrivals.
2. Create a billing queue.
3. Allocate the billing counter using SEIZE.
4. Simulate billing time using ADVANCE.
5. Release the billing counter.
6. Simulate customer exit.
7. Run the simulation for 480 minutes.

8. Analyze queue length and billing counter utilization.

Conclusion

The simulation evaluates supermarket billing efficiency and helps determine whether additional counters are needed.

8. Simulation of Dump Truck using Time Advance Algorithm

Aim

To simulate dump truck operations using the time advance algorithm and analyze cycle time, waiting time, and loader utilization.

Theory

A dump truck system is commonly used in construction and mining industries for transporting materials between loading and dumping sites. The truck performs repeated operations such as loading, traveling, dumping, and returning. The time advance algorithm is used in simulation to move the clock from one event to the next event efficiently. This simulation helps study truck productivity, loader utilization, queue length, and operational delays. It also helps identify bottlenecks in transportation systems. The model is useful for improving scheduling and reducing idle time.

Procedure

1. Create the GPSS model for dump truck operations.
2. Generate the truck transaction at the start of simulation.
3. Create a queue for loading operations.
4. Allocate the loader using the SEIZE block.
5. Simulate loading, transportation, dumping, and return times using ADVANCE.
6. Release the loader after loading is completed.
7. Terminate the transaction after one complete cycle.
8. Run the simulation for 480 minutes.
9. Observe truck utilization, queue length, and average cycle time.
10. Record the number of completed trips and waiting times.
11. Analyze the overall performance of the dump truck system.

Conclusion

The simulation demonstrates efficient dump truck operations using the time advance algorithm. It helps identify delays, waiting times, and loader utilization in transportation systems. The experiment shows how simulation can improve operational planning and resource allocation. It also helps increase productivity by reducing transportation bottlenecks.

9. Probability Distributions for Discrete Events

Aim

To simulate discrete probability distributions using GPSS and study random event behavior.

Theory

Discrete probability distributions are used when outcomes are finite and countable. Each possible outcome has a fixed probability of occurrence. GPSS uses probability-based routing to simulate events such as customer types, machine failures, or item demand. This experiment helps understand how probability distributions influence decision-making systems and event occurrences. Discrete distributions are widely used in queueing and inventory systems. They help represent random behavior in practical applications.

Procedure

1. Generate customer arrivals using the GENERATE block.
2. Use TRANSFER blocks to assign probabilities to different outcomes.
3. Create separate paths for each customer or event type.
4. Simulate activity time using ADVANCE blocks.
5. Terminate transactions after processing.
6. Run the simulation for the specified time period.
7. Observe the number of events generated for each probability category.
8. Compare observed probabilities with expected probabilities.
9. Record the output statistics generated by GPSS.
10. Analyze whether the probability distribution behaves correctly.

Conclusion

The experiment demonstrates how discrete probability distributions are used to model random events in simulation systems. The observed results closely match the assigned probabilities. The simulation verifies the correctness of probability-based routing in GPSS. It also helps understand the role of probabilities in decision-making systems.

10. Probability Distributions for Continuous Events

Aim

To simulate continuous probability distributions and analyze variability in arrival and service times.

Theory

Continuous probability distributions represent variables that can take any value within a range. Exponential distributions are commonly used for arrival patterns, while Normal distributions are used for service times. These distributions help model real-world uncertainty and variability in simulation systems. Continuous distributions improve the realism and accuracy of simulation models. They are widely used in service and manufacturing systems. Simulation with continuous distributions helps predict system performance under uncertain conditions.

Procedure

1. Generate customer arrivals using an exponential distribution.
2. Create a waiting queue using the QUEUE block.
3. Allocate the server using SEIZE.
4. Simulate service time using a normal distribution in ADVANCE.
5. Release the server after service completion.
6. Terminate the customer transaction.
7. Run the simulation for 480 minutes.
8. Observe queue length, waiting time, and server utilization.
9. Record simulation statistics from the output report.
10. Compare system behavior under different probability conditions.

Conclusion

The experiment shows how continuous probability distributions can model real-world random behavior effectively. It helps analyze system performance under variable operating conditions. The simulation demonstrates the importance of variability in arrival and service processes. It also helps improve understanding of stochastic systems.

11. Random Number Generation

Aim

To generate pseudo-random numbers using GPSS and understand their importance in simulation.

Theory

Random numbers are essential for representing uncertainty and randomness in simulation systems. GPSS generates pseudo-random numbers using mathematical algorithms. These numbers are generally uniformly distributed between 0 and 1 and are used for generating arrivals, service times, and probabilities. Random number generation forms the basis of stochastic simulation models. Accurate random numbers improve the reliability of simulation results. They are widely used in Monte Carlo and queueing simulations.

Procedure

1. Initialize the simulation transaction.
2. Use ASSIGN to generate random numbers.
3. Store generated numbers in parameters.
4. Display random numbers using the PRINT block.
5. Repeat the process using LOOP statements.
6. Generate the required number of random values.
7. Terminate the simulation after completion.
8. Observe the generated pseudo-random numbers.
9. Record the generated values for analysis.
10. Verify whether the values are uniformly distributed.

Conclusion

The experiment demonstrates the generation and use of random numbers in simulation. Random numbers are important for modeling probabilistic and uncertain systems. The generated values help simulate real-world randomness effectively. This experiment forms the foundation for advanced simulation techniques.

12. Chi-Square Test to Test Data

Aim

To perform the Chi-Square goodness-of-fit test and verify whether observed data follows an expected probability distribution.

Theory

The Chi-Square test is a statistical method used to compare observed frequencies with expected frequencies. It helps determine whether generated data follows a theoretical distribution such as Uniform distribution. The test is widely used in simulation to validate random number generators and probability distributions. A lower Chi-Square value indicates that the observed data fits the expected distribution. It is commonly used in statistical analysis and hypothesis testing. The test helps improve the accuracy and validity of simulation models.

Procedure

1. Generate random numbers using a simulation method.
2. Divide the data into equal intervals or bins.
3. Count the observed frequencies for each interval.
4. Calculate expected frequencies.
5. Compute the Chi-Square statistic using the formula.
6. Find the critical value from the Chi-Square table.
7. Compare calculated and critical values.
8. Accept or reject the null hypothesis.
9. Record the final test result.
10. Analyze whether the generated data follows the expected distribution.

Conclusion

The Chi-Square test helps verify whether simulation-generated data follows the expected probability distribution. It is useful for validating simulation assumptions and randomness. The experiment improves understanding of statistical testing methods. It also helps ensure reliability of simulation-generated data.

13. K-S Test to Test Data

Aim

To perform the Kolmogorov–Smirnov (K-S) test and determine whether sample data follows a theoretical probability distribution.

Theory

The K-S test is a non-parametric statistical test used to compare sample data with a theoretical distribution. It measures the maximum difference between empirical cumulative distribution and theoretical cumulative distribution. The test is commonly used in simulation studies to validate random number distributions. A smaller D value indicates that the sample data fits the expected distribution. The K-S test is useful for continuous probability distributions. It helps evaluate the accuracy of generated simulation data.

Procedure

1. Generate or collect sample random data.
2. Arrange the data in ascending order.
3. Calculate empirical cumulative frequencies.
4. Calculate theoretical cumulative frequencies.
5. Find the maximum difference between the two values.
6. Compute the K-S statistic.
7. Compare the result with the critical value.
8. Accept or reject the null hypothesis.
9. Record the observed and expected cumulative values.
10. Analyze whether the sample follows the theoretical distribution.

Conclusion

The K-S test is useful for validating simulation-generated data and checking distribution accuracy. It ensures that the generated data follows the required theoretical distribution. The experiment helps understand statistical validation techniques in simulation. It also improves confidence in the reliability of generated data.